

Group name: Car Market Analysis Dashboard



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[Table 1: Group Members]

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1. Introduction:

This project is about using Power BI tools to analyze automotive data, specifically two tables from the Kaggle platform: one with 2022 model year vehicle specifications and another with used car auction pricing.

The project focuses on transforming data into an interactive dashboard that allows users to compare brands, examine prices, and pinpoint elements that affect a car's value to gain a better understanding of the automotive industry. To make the analysis simpler and more beneficial, Power BI tools will also be utilized to clean the data, construct graphs, and add filters and indicators.

All things considered, this project supports improved decision-making when comparing cars and their costs by offering a simplified perspective on automotive data. The project also aligns with Sustainable Development **Goal 9: Industry, Innovation and Infrastructure** by utilizing data analytics, visualization tools, and interactive technologies to support innovation and smarter market analysis within the automotive industry [1].

2. Roles and contributions of team members:

Student Name	Contribution
Ameerah alreshood	Research & Content Writing
Reema alshammari	Data Modeling & Dashboard Design
Feryal kaneeta	Data Collection, Data Cleaning & Preparation
Layan almarshud	Dashboard Implementation, Power BI Features, Conclusion & Future Work
Leena alsaif	Problem Description & Solution Overview

[Table 2: Roles and contributions of team members]

3. Problem Description:

In the automotive market, understanding car prices can be difficult because many factors influence their value. Many users lack simple tools to analyze car data and identify important patterns. Without clear insights into factors such as brand, model, year, and mileage, it becomes difficult to make informed decisions. In addition, the data may contain missing values and inconsistencies, which can reduce the accuracy of the analysis. This creates a need for a clear and interactive way to explore the data and support better decision-making.

4. Overview of Solution:

To address the challenges in understanding and analyzing car prices, this project uses Power BI to develop a clear and interactive dashboard. By presenting key factors such as brand, model, year, and mileage in visual form, users can easily explore the data and recognize patterns that affect car value. These visual insights improve the understanding of price variations and support users in making more accurate and informed decisions in the automotive market.

5. Description of data:

Our data was collected from Kaggle. The first table contains car prices and serves as the main table, while the second table includes car specifications.

The main table consists of 558,837 records and 17 features. Examples of these features include the year, condition, selling price, and additional information such as state, location, type, and others [2].

The second table contains 1,500 records and 13 features, including model, seating capacity, rating, max power, fuel tank capacity, and other related features [3].

6. Data preparation:

The two tables need to be cleaned and prepared before analysis. We started by preparing the car specifications table using a Python notebook. First, the table was loaded, and its shape, data types, and missing values were examined.

Missing values in critical columns such as model and make were handled by removing rows with missing entries, while missing values in the body column were replaced with the most frequent value (mode). After that, the data was standardized by cleaning and formatting text columns, and numeric columns were ensured to have the correct data types.

Next, outliers were detected and handled using the IQR method. Finally, the cleaned table was saved for further analysis.

The car prices table was cleaned and prepared using a Python notebook. First, the table was loaded, and an initial overview was performed to identify the data shape, data types, and missing values.

Unnecessary columns such as Unnamed and saledate were removed as they do not contribute

to the analysis. Missing values in categorical features such as trim, body, color, and interior were handled by replacing them with the most frequent value (mode). For the transmission_type column, rows with missing values were removed due to the large number of missing entries.

Text columns such as make, model, and body were standardized by converting them to lowercase and removing extra spaces to ensure consistency. Numerical columns such as

odometer and condition were converted to numeric format, and missing values were filled using the median.

Outliers in the sellingprice column were detected and removed using the IQR method to improve data quality. Finally, rows with missing values in critical columns such as make and model were removed.

The cleaned table was then saved for further analysis and visualization.

The Car_Specifications table preprocessing :

1- Load and data overview:

```
import pandas as pd
import numpy as np

# Load dataset
df2 = pd.read_csv("Car_Specifications.csv")

# 1. Overview

print("Shape:", df2.shape)
print("\nData Types:\n", df2.dtypes)
print("\nMissing Values:\n", df2.isnull().sum())
```

output:

```
Shape: (2000, 13)

Data Types:
make          str
model         str
seating_capacity  int64
transmission_type  str
fuel_tank_capacity  int64
body          str
rating        float64
starting_price  int64
ending_price   int64
max_torque_nm  int64
max_torque_rpm  int64
max_power_bhp  int64
max_power_rpm  int64
dtypes: object

Missing Values:
make          9
model        10
seating_capacity  0
transmission_type  0
fuel_tank_capacity  0
body          37
rating         0
starting_price  0
ending_price   0
max_torque_nm  0
max_torque_rpm  0
max_power_bhp  0
max_power_rpm  0
dtype: int64
```

Figure 1: Car Specifications table Loading and Overview

2- Handle missing value:

```
#1 remove rows with missing make or model
df = df2.dropna(subset=["make", "model"])

# fill missing body with the most repeated value
df["body"] = df2["body"].fillna(df["body"].mode()[0])

# check again
df.isnull().sum()
```

```
make          0
model         0
seating_capacity  0
transmission_type  0
fuel_tank_capacity  0
body          0
rating         0
starting_price  0
ending_price   0
max_torque_nm  0
max_torque_rpm  0
max_power_bhp  0
max_power_rpm  0
dtype: int64
```

Figure 2: Handling Missing Values in the Car Specifications table

3- Standardize text columns and Ensure numeric columns:

```
#2 Standardize text columns
df2["make"] = df2["make"].str.lower().str.strip()
df2["model"] = df2["model"].str.lower().str.strip()
df2["body"] = df2["body"].str.lower().str.strip()

[11]:

# 3. Ensure numeric columns

num_cols = [
    "seating_capacity", "fuel_tank_capacity", "rating",
    "starting_price", "ending_price",
    "max_torque_nm", "max_torque_rpm",
    "max_power_bhp", "max_power_rp"
]

for col in num_cols:
    df2[col] = pd.to_numeric(df2[col], errors="coerce")
```

Figure 3: Standardizing Text Columns in the Car Specifications table

4- Outlier Detection (IQR):

```
# 4. Outlier Detection (IQR)

def detect_outliers(df, column):
    Q1 = df[column].quantile(0.25)
    Q3 = df[column].quantile(0.75)
    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    outliers = df[(df[column] < lower) | (df[column] > upper)]
    print(f'{column} outliers:', outliers.shape[0])

    return lower, upper
```

Figure 4: Ensuring Numeric Columns in the Car Specifications table

5- For further analysis.

```
df2.to_csv("car_Specifications_BI.csv", index=False)
```

Figure 5: Saving the Cleaned Car Specifications Table for Further Analysis

The Car_prices table preprocessing :

1- Load and data overview:

output:

```
import pandas as pd
import numpy as np

# Load dataset
df = pd.read_csv("car_prices.csv", on_bad_lines="skip")

# 1. Basic Overview
print("Shape:", df.shape)
print("\nData Types:\n", df.dtypes)
print("\nMissing Values:\n", df.isnull().sum())
```

```
Shape: (558837, 17)

Data Types:
year          int64
make          str
model         str
trim          str
body          str
transmission_type str
vin           str
state         str
condition     object
odometer      float64
color         str
interior      str
seller        str
mmr           object
sellingprice  int64
saledate      str
Unnamed: 16   str
dtype: object

Missing Values:
year          0
make         10301
model        10399
trim         10651
body         13195
transmission_type 65353
vin           4
state         0
condition     11794
odometer      94
color         749
interior      749
seller         0
mmr           0
sellingprice  0
saledate      0
Unnamed: 16   558811
dtype: int64
```

Figure 6: Car Prices Table Loading and Data Overview

2- Handle missing value:

```
# 1. Drop useless column

df.drop(columns=["Unnamed: 16"], inplace=True)
df.drop(columns=["saledate"], inplace=True)

# [3]:

#2 Handle remaining missing values

# Fill trim, body, color, interior with mode
cols_mode = ["trim", "body", "color", "interior"]

for col in cols_mode:
    df[col] = df[col].fillna(df[col].mode()[0])

# Fill missing transmission values
df = df.dropna(subset=["transmission_type"])
df = df.dropna(subset=["vin"])
```

Figure 7: Handling Missing Values in the Car Prices Table

3- Standardize text columns and Fix Numerical Columns

```
#3 Standardize text columns
df["make"] = df["make"].str.lower().str.strip()
df["model"] = df["model"].str.lower().str.strip()
df["body"] = df["body"].str.lower().str.strip()

[13]:

#4 Fix Numerical Columns Properly

# Convert columns to numeric (force errors to NaN)
df["odometer"] = pd.to_numeric(df["odometer"], errors="coerce")
df["condition"] = pd.to_numeric(df["condition"], errors="coerce")

# Fill missing values WITHOUT inplace (recommended)
df["odometer"] = df["odometer"].fillna(df["odometer"].median())
df["condition"] = df["condition"].fillna(df["condition"].median())
```

Figure 8: Standardizing Text Columns and Fixing Numerical Columns in the Car Prices Table

4- Remove outliers and Drop rows with missing critical columns

```
#5 Remove outliers using IQR
Q1 = df["sellingprice"].quantile(0.25)
Q3 = df["sellingprice"].quantile(0.75)
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

df = df[(df["sellingprice"] >= lower) & (df["sellingprice"] <= upper)]

[16]:

# Drop rows with missing critical columns again
df = df.dropna(subset=["make", "model"])
```

Figure 9: Removing Outliers and Handling Missing Critical Columns in the Car Prices Table

5- Load the new cleaned data:

```
df.to_csv("car_prices_BI.csv", index=False)
```

Figure 10: Saving the Cleaned Car Prices Table for Further Analysis

7. Visual Model:

To establish a clear analytical structure and ensure efficient data processing within this project, the table was organized into three tables. This modeling approach enabled the creation of accurate relationships, minimized redundancy, and supported interactive reporting in Power BI. All tables are connected through a uniquely generated field, Car_Key, which serves as the central link across the model.

The explanation of each table:

1. car_prices_BI Table

- This table functions as the main fact table and contains detailed information about car listings and market-related attributes.
- It includes fields such as body type, exterior color, vehicle condition, odometer readings, selling price, state, seller information, and transmission type.
- Because it holds the transactional and market-driven data, this table forms the foundation for analyzing pricing patterns, identifying market trends, and comparing car categories based on real-world sales behavior.
- Most pricing visuals, KPIs, and distribution charts in the dashboard rely on this table [2].

2. car_Specifications_BI Table

- This dimension table contains the technical and mechanical specifications of each car.
- It includes attributes such as horsepower (BHP), engine speed at maximum power (RPM), torque, fuel tank capacity, seating capacity, and overall rating.
- The table supports performance-based analysis and enables comparisons between different car models in terms of power, efficiency, and design characteristics.
- It also helps identify relationships between technical specifications and market outcomes, such as how horsepower or torque may influence selling price or customer ratings [3].

3. car_prices_unique Table

- This table serves as a bridge table designed to resolve duplication issues and ensure accurate relationships between the other tables.
- It contains the Car_Key field, which uniquely identifies each car record and eliminates repeated values that previously caused Many-to-Many relationships.
- By linking both the pricing and specifications tables through this unique key, the model achieves clean One-to-Many relationships, enabling consistent filtering, correct aggregation, and reliable interaction across all visuals.
- This table plays a critical role in maintaining data integrity and ensuring that slicers affect all visuals uniformly.

By structuring the table into these three interconnected tables, the data model became more organized, scalable, and analytically robust. This design supports precise filtering, enhances the clarity of relationships between performance and pricing, and enables the dashboard to deliver meaningful and accurate insights across all visual components.

8. Dashboard design overview:

8.1. Dashboard Structure and Key Visual Elements

The dashboard is designed as a single interactive page that brings together the most important visuals needed to analyze the car market dataset. Each visual highlights a specific aspect of performance, pricing, or distribution, and all elements respond dynamically to the slicers.

- **Total Cars:** A KPI card showing the total number of cars in the dataset, giving an immediate overview of market size.
- **Performance KPIs:** Two KPI cards display the highest horsepower (400 BHP) and maximum engine speed (7K RPM), summarizing the top performance values in the dataset.
- **Average Rating by Make:** A bar chart comparing the average rating across car brands to identify which manufacturers receive higher user evaluations.
- **Average Horsepower by Make:** A bar chart showing the average horsepower for each brand, supporting performance comparison between manufacturers.
- **Car Distribution by State:** A map visual illustrating how cars are distributed across states, helping identify regional patterns in availability.
- **Average Selling Price by Make:** A bar chart showing the average selling price for each brand, highlighting pricing differences between economy and premium brands.
- **Total Cars by Make:** A donut chart representing the share of each brand in the dataset, giving a clear view of brand dominance.
- **Average Mileage and Maximum Selling Price:** A combined chart comparing average mileage (Km) and maximum selling price (\$) for each brand to explore the relationship between usage and value.

This layout provides a clear, organized, and interactive view of the dataset, allowing users to explore pricing, performance, and distribution patterns efficiently.

8.2. Design and Aesthetic Choices

- **Color Theme:** A light background combined with a palette of soft blue and gray tones was selected to create a clean, professional, and visually balanced interface. The color harmony enhances readability and gives the dashboard a modern, data-driven appearance suitable for academic presentation.
- **Interactivity:** The dashboard integrates slicers and filters that allow users to explore data dynamically by car make & body type. This interactive design supports detailed comparisons and helps users focus on specific market segments.
- **Visual Consistency:** All charts and visuals follow a unified style with consistent labeling, font sizes, and legend placement. This ensures clarity and coherence across both pages of the dashboard, making the analytical flow intuitive and visually appealing.
- **Ease of Use:** The dashboard was designed with simplicity and accessibility in mind. Even users with limited Power BI experience can navigate easily, interpret results, and interact with visuals without confusion.

This design approach ensures that the dashboard communicates insights effectively, supports analytical decision-making, and maintains a professional aesthetic aligned with academic standards.



Figure 11: Car Market Analysis Dashboard – Pricing and Market Distribution Page



9. Implementation:

9.1 Custom Calculations measures:

A- Measure **car_prices_BI** Table

1. **Average Mileage:**
This measure was created to calculate the average mileage of all cars in the dataset. It helps analyze the overall usage and driving distance of vehicles.
2. **Average Selling Price:**
This measure calculates the average selling price of cars. It was used to compare car prices across different brands and categories.
3. **Max Selling Price:**
This measure identifies the maximum selling price in the dataset. It helps determine the highest-priced car available in the market.
4. **Min Selling Price:**
This measure calculates the minimum selling price of cars. It provides insight into the lowest-priced vehicles in the dataset.
5. **Price Difference:**
This measure was created to calculate the difference between car prices for comparative analysis and better understanding of price variations.
6. **Total Cars:**
This measure counts the total number of cars included in the dataset. It was used to summarize the dataset size and support dashboard analysis.

B- Measure **car_Specifications_BI** Table

1. **Average Power per Make:**
This measure was created to calculate the average horsepower for each car brand. It helps compare vehicle performance across different manufacturers.
2. **Average Rating:**
This measure calculates the average car rating in the dataset. It was used to evaluate overall customer satisfaction and compare ratings between brands.

9.2 Dashboard Layout:

The dashboard title is centered to maintain focus on the main analysis topic. Interactive features such as slicers are placed on the left side, allowing users to filter data and explore specific car brands and categories easily. KPI cards are positioned clearly to provide quick insights into important metrics such as selling prices and car performance. Visualizations are arranged centrally to support fast and effective analysis while maintaining a clear and organized dashboard layout. Figure 2 and Figure 3

9.3 Visuals

The dashboard visuals were carefully selected to represent the data effectively, identify patterns, and generate meaningful insights. Different visualization types such as bar charts, KPI cards, donut charts, and maps were used to support comparative analysis and improve data interpretation.

9.3.1 Slicer:

- Interactive slicers were added to allow users to filter and explore the dashboard data easily. Figure 13.
- Users can filter the dashboard based on car make and body type to compare different vehicle categories and analyze specific brands more effectively.

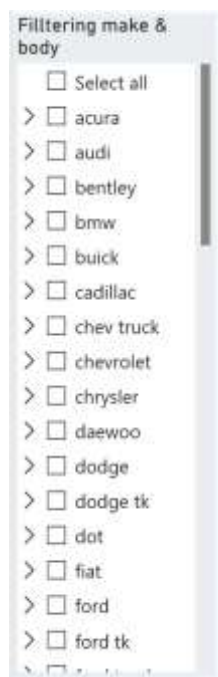


Figure 13: Slicer

9.3.2 Map:

- A map visualization was used to display the geographical distribution of cars across different states and regions. Figure 14.
- The map allows users to identify where cars are distributed and observe the concentration of vehicles in different locations.
- The blue markers represent car locations across multiple states, helping users gain quick geographical insights and understand regional distribution patterns within the dataset.



Figure 14:Map

9.3.3 Clustered Bar Chart:

- A clustered bar chart was used to compare the average selling prices of different car brands. Figure 15.
- The chart helps identify the brands with the highest and lowest average selling prices, allowing users to compare market values across manufacturers. The results show that brands such as Maserati, Bentley, and Ram have higher average selling prices compared to other brands in the dataset.



Figure 15: Clustered bar chart

9.3.4 Clustered Column Charts

- Clustered column charts were used to compare multiple performance and pricing metrics across different car brands. Figures 16, 17, and 18.
- The Average Horsepower chart highlights differences in engine performance between manufacturers, where brands such as BMW, Nissan, and Infiniti show higher horsepower values compared to other brands.
- The Average Rating chart was used to compare customer ratings across car brands. The results indicate that most manufacturers received relatively high and similar ratings with only slight variations.
- The Average Mileage and Max Selling Price chart was created to analyze the relationship between vehicle usage and pricing. Some brands show higher mileage with lower prices, while others maintain higher selling prices despite lower mileage, providing insights into market value and performance differences between manufacturers.



Figure 16: Clustered Column Chart – Average Mileage and Max Selling Price by Make

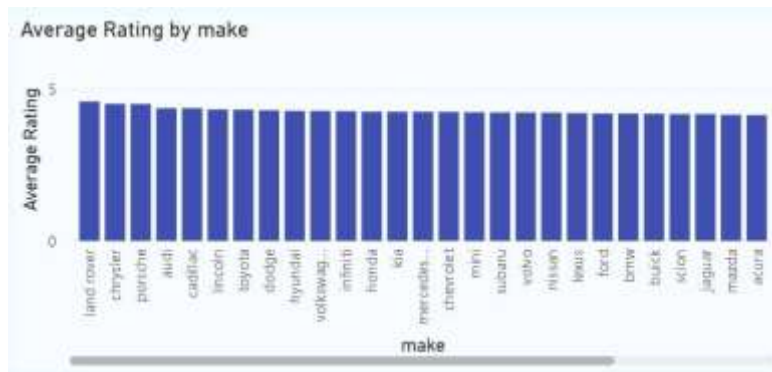


Figure 17: Clustered Column Chart – Average Rating by Make

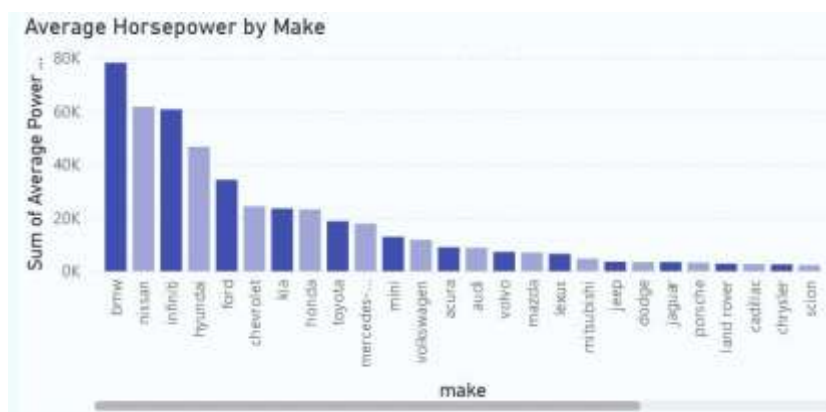


Figure 18: Clustered Column Chart – Average Horsepower by Make

9.3.5 KPI Cards:

- KPI cards were used to display important summary metrics and provide quick insights into car pricing and vehicle performance. Figures 19 and 20.
- The first KPI cards present the minimum and maximum selling prices in the table, helping users quickly identify the price range of vehicles.
- Additional KPI cards were created to display the maximum horsepower (BHP) and the engine speed at maximum power (RPM), allowing users to analyze overall vehicle performance and engine capability efficiently.



Figure 19: KPI Cards – Minimum and Maximum Selling Prices



Figure 20: KPI Cards – Maximum Horsepower and Engine Speed at Maximum Power

9.3.6 Donut Chart:

- A donut chart was used to display the distribution of cars by make and show the proportion of each car brand within the dataset. Figure 21.
- The chart simplifies the comparison between manufacturers by representing each brand as a percentage of the total number of cars. The values and percentages displayed around the chart indicate the contribution of each brand to the overall dataset.
- The results show that brands such as Ford, Chevrolet, and Nissan represent a larger share of the dataset compared to other manufacturers, providing insights into the most common car brands included in the analysis.

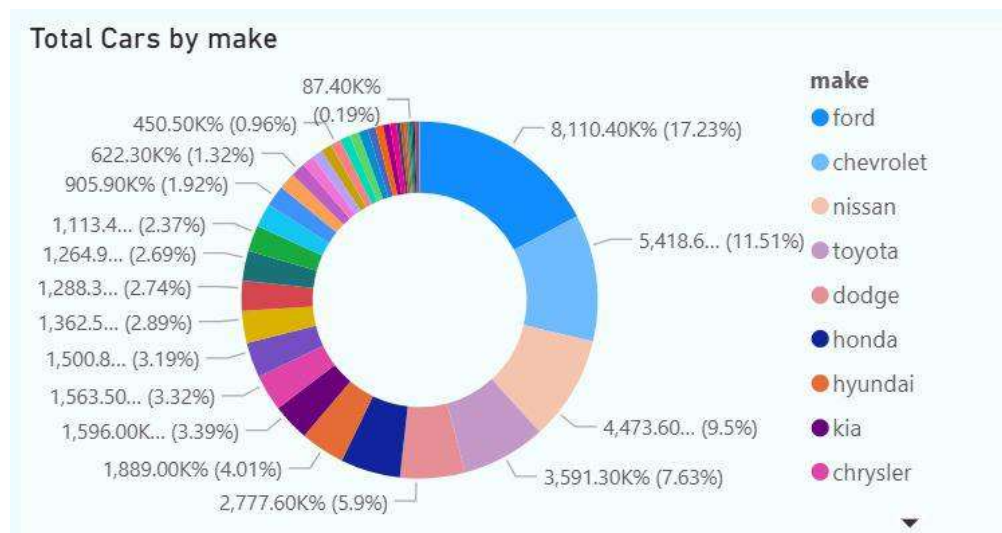


Figure 21: Donut chart

10. Power BI features used:

During the development of the Car Market Analysis Dashboard, several Power BI features were utilized to enhance interactivity and support data analysis:

- **Filters and Slicers:** Enabled dynamic filtering by car make and body type, allowing users to focus on specific segments. Figure 13
- **KPI Cards:** Used to display important metrics such as maximum horsepower (BHP), engine speed (RPM), and selling prices. Figures 19 and 20.
- **Charts and Visuals:** Implemented clustered column charts, and donut charts to compare average ratings, horsepower, selling prices, and total car counts. Figures 16, 17, and 18,21.
- **Map Visualization:** Applied to display car distribution by state, adding a geographical dimension to the analysis. Figure 14.
- **Data Modeling:** Relationships were established between tables to ensure accurate filtering and consistent data aggregation. Figure 22.

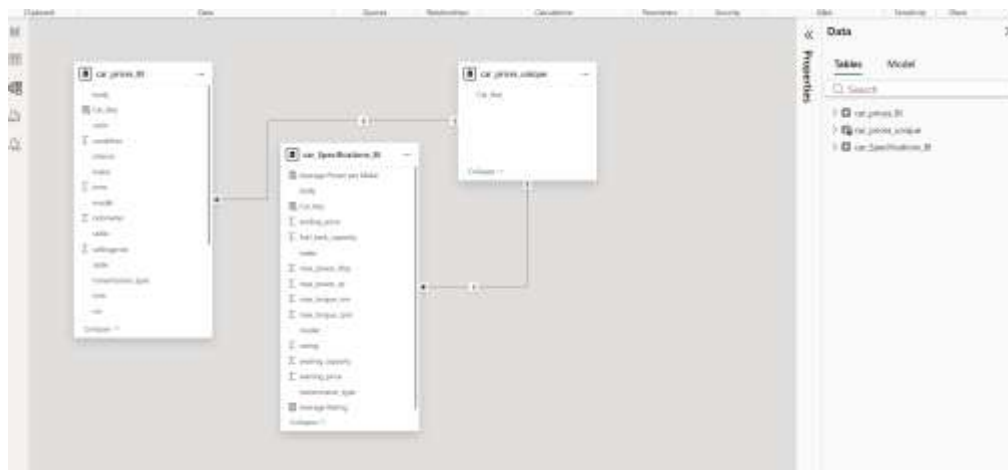


Figure 22: Relationship Model Between Tables

11. Problems faced:

1. Data Modeling Challenges

One of the main challenges was establishing correct relationships between the datasets because the Car_Key column contained duplicate values in multiple tables. This initially caused many-to-many relationship issues and inconsistent filtering behavior across visuals. To solve this problem, an additional table containing unique Car_Key values was created to support proper one-to-many relationships and improve data modeling accuracy.

2. Data Cleaning and Preparation Issues

The datasets contained duplicate values, missing data, and inconsistent text formatting in columns such as make, model, and car_key. Additional cleaning and preprocessing steps were required to improve data quality and ensure accurate analysis.

3. Slicer and Interaction Configuration

Configuring slicers and interactive filters was challenging at the beginning, especially when adjusting multi-select settings and ensuring that filters affected all dashboard visuals correctly.

4. Dashboard Layout and Visual Design

Maintaining a consistent dashboard layout required several adjustments to chart sizes, alignment, colors, and spacing to improve readability and create a professional appearance.

5. Data Formatting Issues

Some numerical values initially appeared in incorrect formats, such as percentages instead of currency values. Formatting settings were adjusted to ensure consistency across all dashboard visuals.

12. Limitations:

Despite the effectiveness of the dashboard, several limitations were identified during the project. Some car brands contained a limited number of records compared to others, which may affect the accuracy of comparisons and overall analysis. In addition, the datasets did not include certain important features such as fuel efficiency, accident history, or maintenance records, which could provide deeper insights into vehicle performance and pricing. Some visualizations were also limited by the available data structure and Power BI visual settings. Furthermore, the analysis depends entirely on the quality and completeness of the original datasets, meaning that any missing or inconsistent values may influence the final results and insights.

13. Future Work:

For future work, the dashboard could be expanded by integrating additional datasets related to fuel efficiency, maintenance history, accident records, and customer reviews to provide deeper insights into vehicle performance and market value. More advanced analytical techniques such as predictive analytics and machine learning models could also be applied to predict car prices and identify market trends more accurately. In addition, the dashboard could be enhanced with more interactive features, real-time data updates, and advanced filtering options to improve the user experience and support better decision-making.

14. Conclusion:

In our project, we developed an interactive Car Market Analysis Dashboard using Power BI to analyze car prices, specifications, and performance across different manufacturers. The dashboard helped transform raw data into meaningful visual insights through the use of KPI cards, charts, maps, slicers, and data modeling techniques.

The analysis provided useful insights into selling prices, horsepower, mileage, ratings, and geographical distribution of vehicles. By applying data cleaning, relationship modeling, and custom calculations, the dashboard became more accurate, interactive, and easier to explore. Overall, our project demonstrates how Power BI can be used to support data analysis and improve decision-making through effective visualization and interactive dashboard design.

15. References:

- [1] United Nations, "Goal 9: Industry, Innovation and Infrastructure," Sustainable Development Goals, 2023. [Online]. Available: <https://sdgs.un.org/goals/goal9>.
- [2] B. Tunguz, "Used Car Auction Prices Dataset," Kaggle, 2022. [Online]. Available: https://www.kaggle.com/datasets/tunguz/used-car-auction-prices?select=car_prices.csv.
- [3] T. Tr1gg3rtrash, "Cars 2022 Dataset," Kaggle, 2022. [Online]. Available: <https://www.kaggle.com/datasets/tr1gg3rtrash/cars-2022-dataset>.